Power + HV bias

File: power_hv.kicad_sch

4x SiPM analog front end

File: afe.kicad_sch

FPGA + DAC + sensors + comms

File: digital.kicad_sch

4–channel SiPM cosmic–ray muon telescope front end

Topology per simulation report 2026–06–10:

OPA858 TIA (Rf=2k, Cf=2.2p, 24.4mV/p.e., 52MHz) –> TLV3601 –> iCE40UP5K

TPS61170 boost -30V (MicroFC–30035) + pi filter + DAC trim + ADC readback

nRF9151 on carrier headers: housekeeping, I2C (DAC7578/BME280), LTE–M uplink

LC76G GNSS module header: PPS to FPGA for GPS–disciplined timestamps

Symbols: CDFER JLCPCB–Kicad–Library (passives/BAV99/BME280/W25Q128/AMS1117,

LCSC numbers embedded) + datasheet–verified customs (see README + LCSC fields).

BEFORE ORDERING: run ERC, verify items marked VERIFY, assign extended–part LCSC IDs.

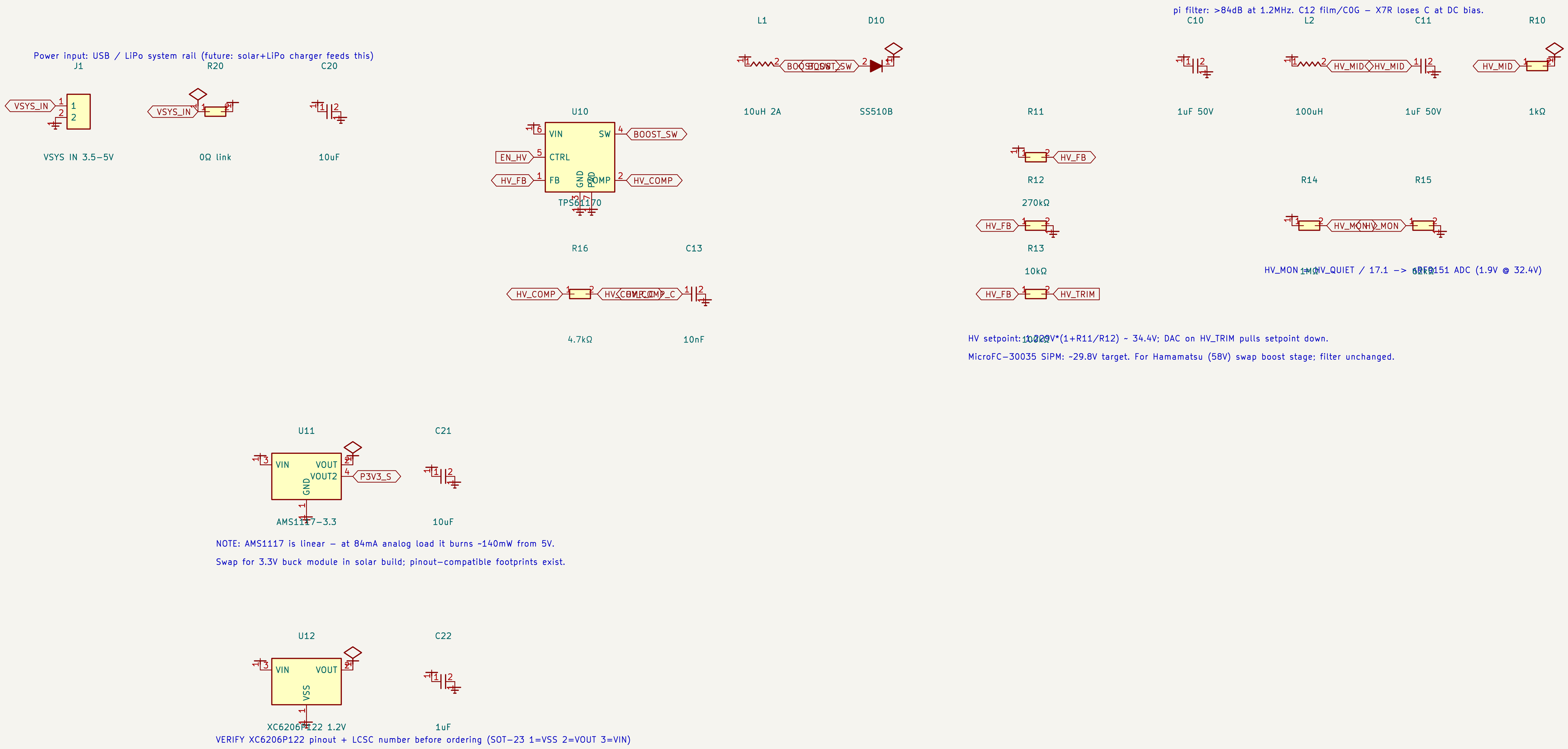
SiPM Muon Telescope

Sheet: /

File: muon_telescope.kicad_sch

Title: SiPM Muon Telescope – root

Size: A3	Date: 2026–06–10	**Rev: A**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/4



Generated: CDFER JLCPCB lib + verified datasheet pinouts

SiPM Muon Telescope

Sheet: /Power + HV bias/

File: power_hv.kicad_sch

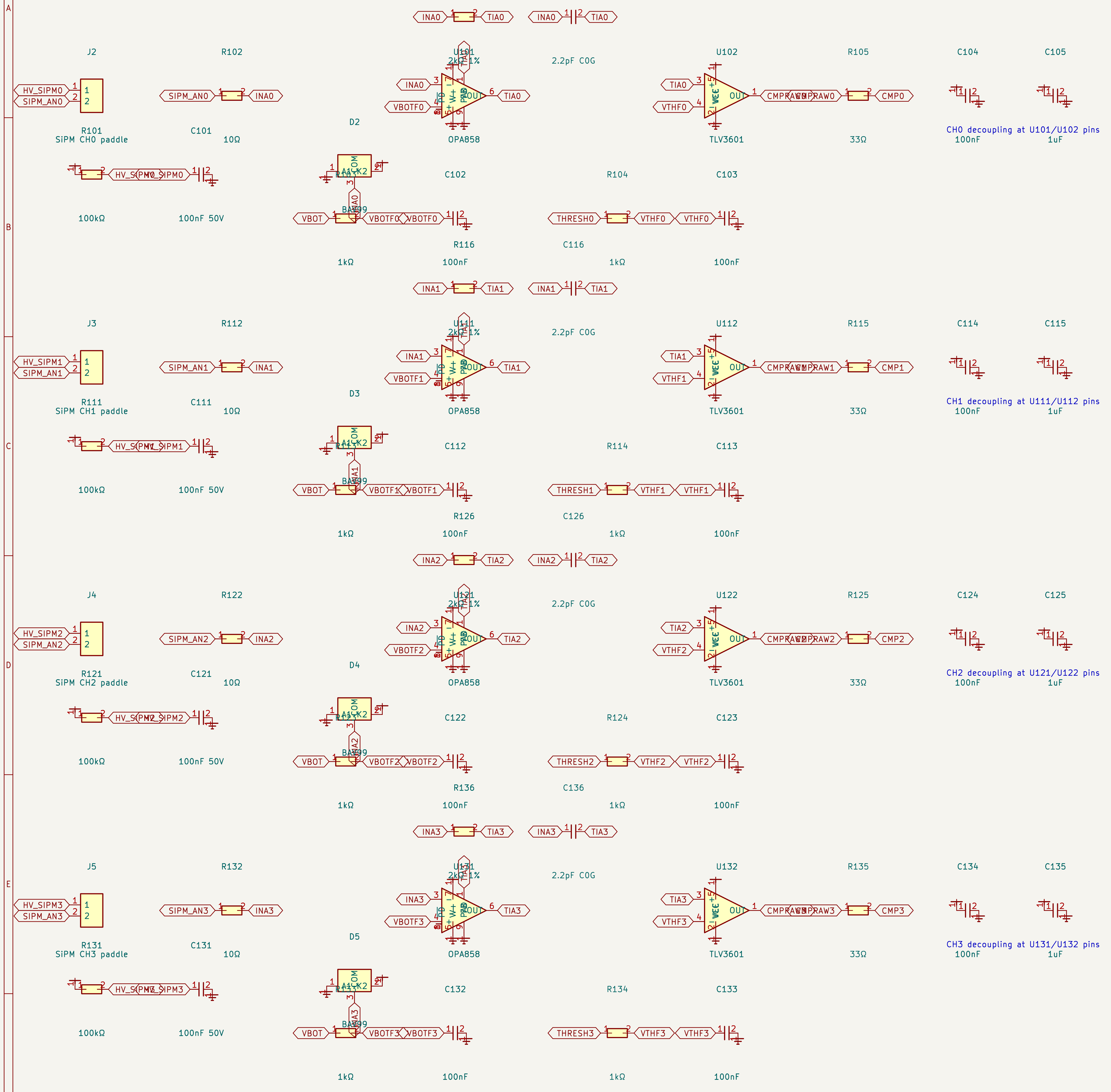
Title: Power input, 3V3/1V2 rails, HV bias generator

Size: A3 | Date: 2026–06–10 | Rev: A

KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 | Id: 2/4

Per channel: DC−coupled anode readout. OPA858 TIA Rf=2k Cf=2.2p −> 24.4mV/p.e., 52MHz.

TLV3601: IN+=TIA_OUT, IN−=THRESH −> active−low CMPx. ToT = 14.6*ln(Npe)−7.5ns.



Generated: CDFER JLCPCB lib + verified datasheet pinouts

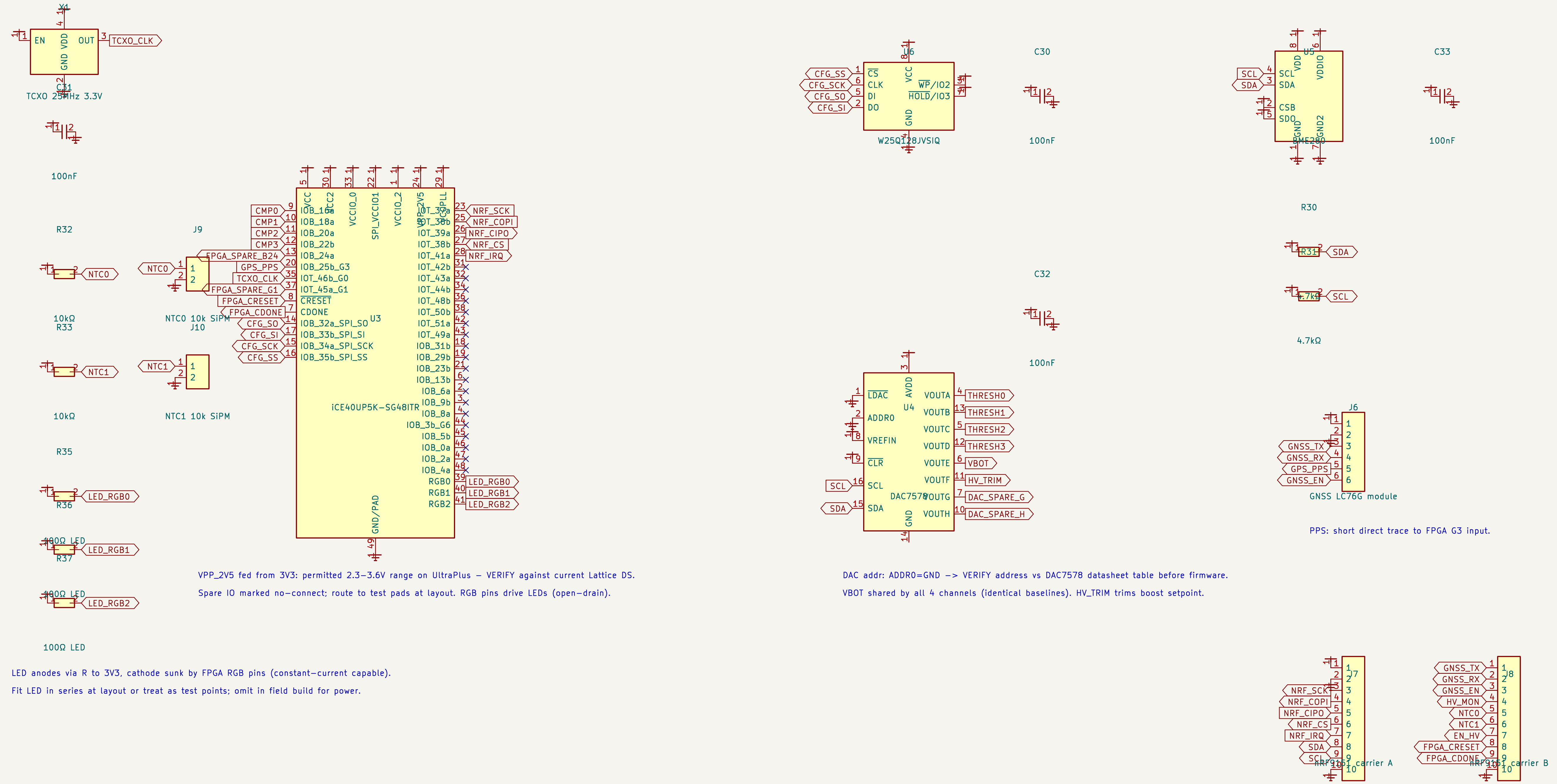
SIPM Muon Telescope

Sheet: /4x SiPM analog front end/

File: afe.kicad_sch

Title: Analog front end: 4x SiPM TIA + comparator (per sim report 2026-06-

Size: A3	Date: 2026-06-10	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4



VPP_2V5 fed from 3V3: permitted 2.3–3.6V range on UltraPlus – VERIFY against current Lattice DS.

Spare IO marked no–connect; route to test pads at layout. RGB pins drive LEDs (open–drain).

DAC addr: ADDR0=GND –> VERIFY address vs DAC7578 datasheet table before firmware.

VBOT shared by all 4 channels (identical baselines). HV_TRIM trims boost setpoint.

PPS: short direct trace to FPGA G3 input.

LED anodes via R to 3V3, cathode sunk by FPGA RGB pins (constant–current capable).

Fit LED in series at layout or treat as test points; omit in field build for power.

Generated: CDFER JLCPCB lib + verified datasheet pinouts

SiPM Muon Telescope

Sheet: /FPGA + DAC + sensors + comms/

File: digital.kicad_sch

Title: iCE40UP5K coincidence engine, DAC, sensors, GNSS + nRF9151 carrier

Size: A3	Date: 2026-06-10	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4